

UNIVERSAL PLASTIC SHREDDER V2.0

Electrical Control & Power System

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The Problem

- Ichor Systems creates large volumes of thick rigid plastic waste (PVC, PP, LDPE) from semiconductor manufacturing.
- Sheets are 0.25–0.5 in. thick, 4–12 in. wide – off-the-shelf shredders are too small or too expensive.
- Our scope: **electrical control + power system**. Mechanical frame and blades come from the ME team.

Requirements

Must:

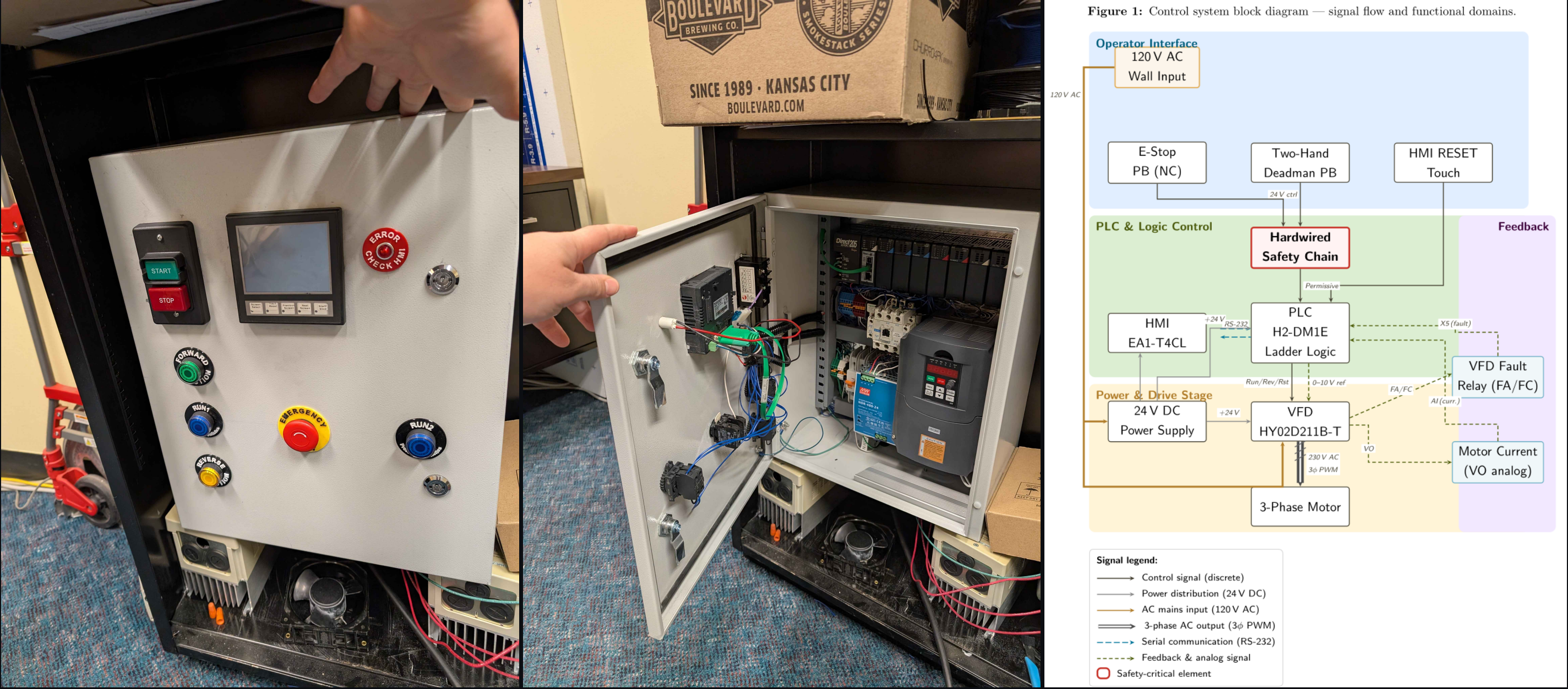
- Shred plastic to ~1 in. pieces.
- Drive a 3–5 HP motor at 70–90 RPM through a controllable VFD.
- Follow NFPA 70 (NEC); include E-stop, overload, and one safety interlock.
- Provide an HMI for operator control and status.
- Stay within the \$1,000 EE-team budget.

Should: status / fault lights, NEMA enclosure, labeled controls.

May: read-only wireless monitoring, data logging, capacitive-touch safety.

System Architecture

- Industrial stack:** AutomationDirect end-to-end. No hobby boards in the control or safety path.
- Power path:** PowerTec 71008 disconnect → Mean Well 24 VDC PSU → SD-N35 contactor → Huanyang VFD → motor.
- Motor:** IronHorse MTC2P-002, 2 HP, 230 V, 1735 RPM, VFD-rated.
- Safety:** E-stop hardwired through the contactor coil, parallel to the PLC.



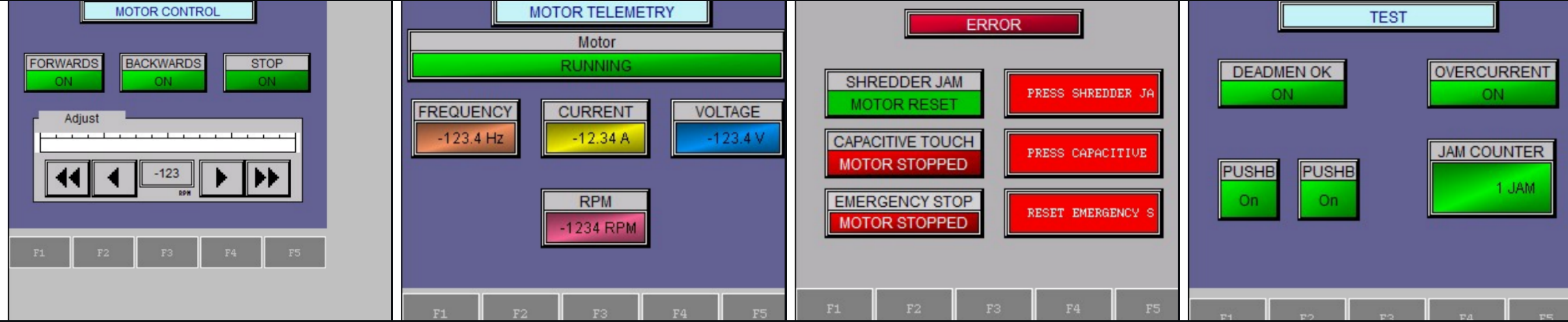
Operator face • panel interior • system block diagram.

PLC Control

- CPU:** DirectLOGIC 205 + H2-DM1E.
- I/O:** 8 DC in, 8 relay out, 8 analog in, 4 RTD in, 0–10 V analog out to the VFD.
- Program:** 31 rungs in Do-more. Deterministic state machine, 3-strike jam recovery, hard-lockout path.

HMI

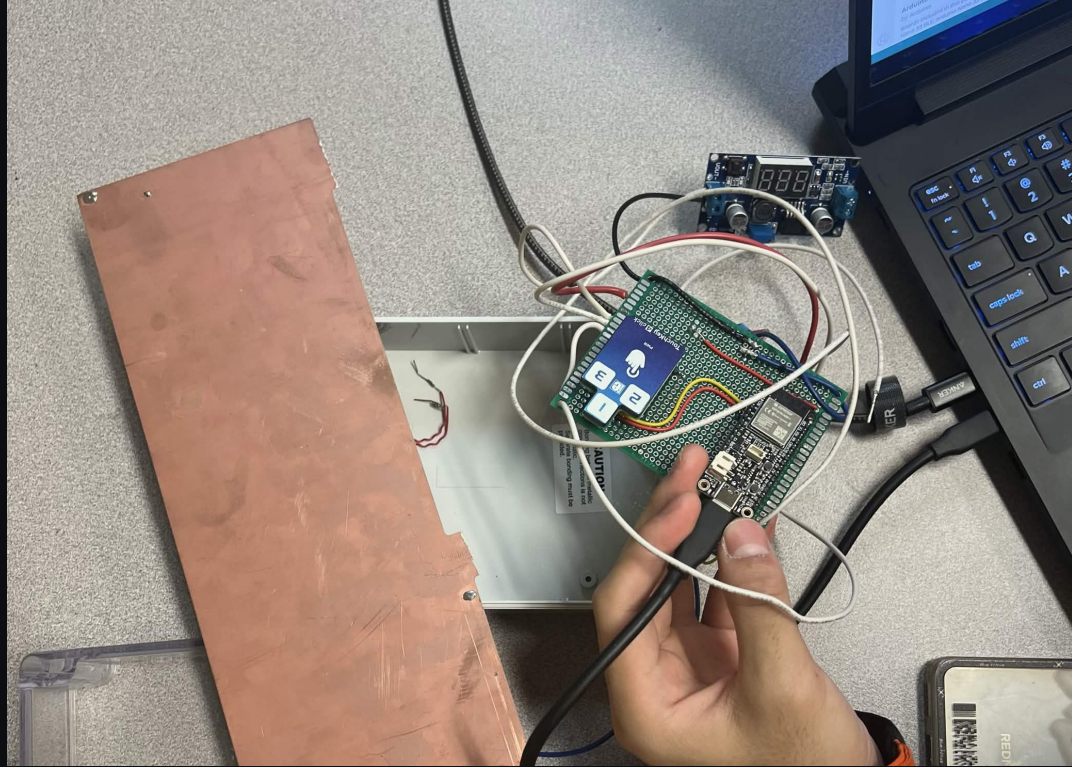
- Panel:** EA1-T4CL C-more Micro-Graphic, 4 in. touch.
- Project:** 4 screens, 27 PLC tags, serial to the CPU.



Motor Control • Motor Telemetry • Error / Alarm • Test.

Capacitive-Touch Safety (CTSI) Module

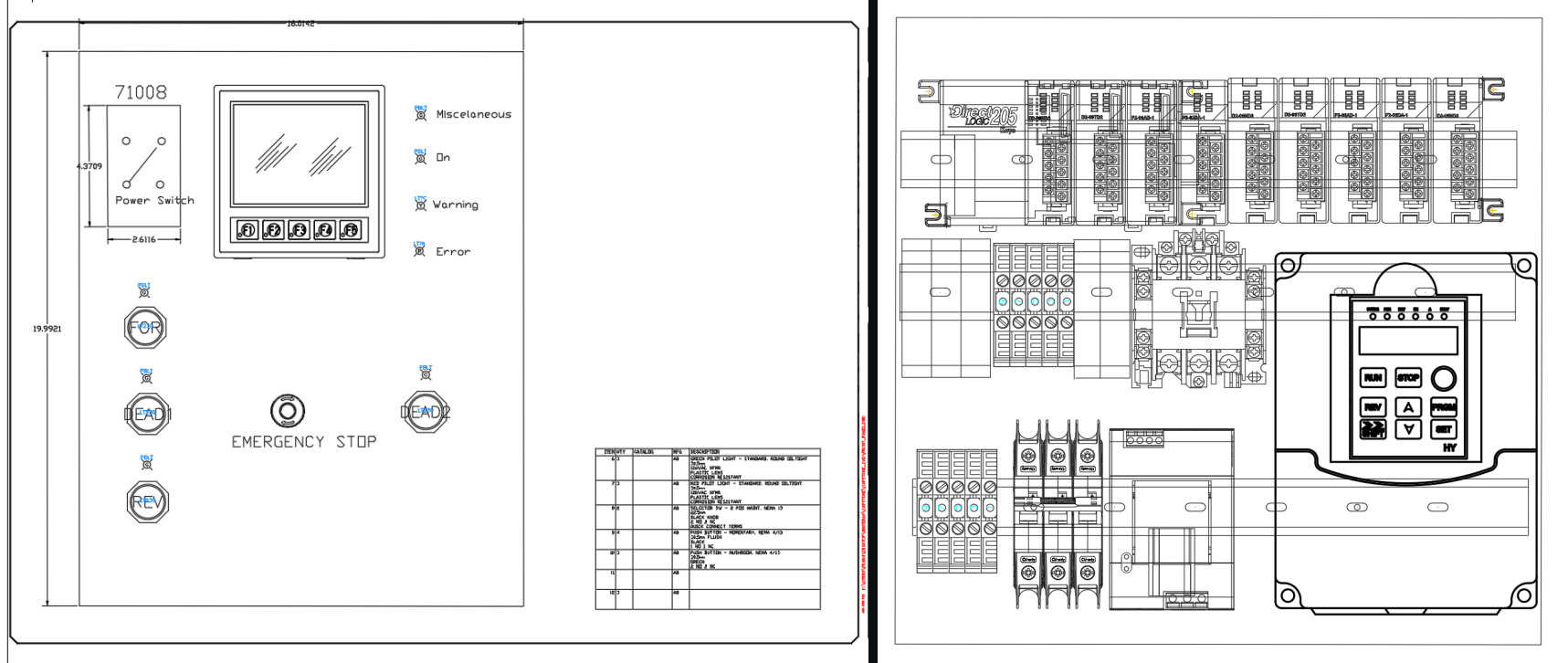
- What:** contact-free operator-presence sensor, in-house build.
- How:** ESP32 reads electrode → optocoupler closes 24 V on PLC X7.
- Power:** panel 24 VDC → LM2596 buck → 3.3 V to ESP32.



ESP32 + LM2596 buck + optocoupler module wired to PLC X7.

On touch (rung 28): motor stops, fault LED on, HMI jumps to ERROR – independent of the deadman / E-stop chain.
Status: bench-validated. Full-machine test pending the ME prototype.

Panel CAD Layout



AutoCAD Electrical: front operator face (left) • back-plate component layout (right).

Safety & Compliance

- E-stop:** Category 0 stop. Eaton E22B1 NC mushroom in series with the SD-N35 contactor coil. Drops the contactor independently of the PLC.
- Two-hand / deadman:** both pushbuttons must be held to enable the motor (PLC X2 AND X3).
- Overcurrent / jam:** VFD over-torque trip ($PD124 = 150\%$, $PD125 = 3\text{ s}$) on PLC X5; ladder runs the 3-strike unjam routine.
- Enclosure:** Vorksmen SEB001 steel, NEMA Type 12.
- Compliance:** NFPA 70/79, NEMA 250 / ICS / MG 1, ISO 12100, 13849-1 (Cat B–1 / PL b–c), 13850, 13851, OSHA 1910.147.

Results

All EE “Must” requirements met – motor control, NFPA-70 design, E-stop + overload + interlock, HMI, and the \$1,000 budget.
Constraint: ME team did not deliver the mechanical prototype, so full-machine integration testing was not possible. All EE validation is bench-only.

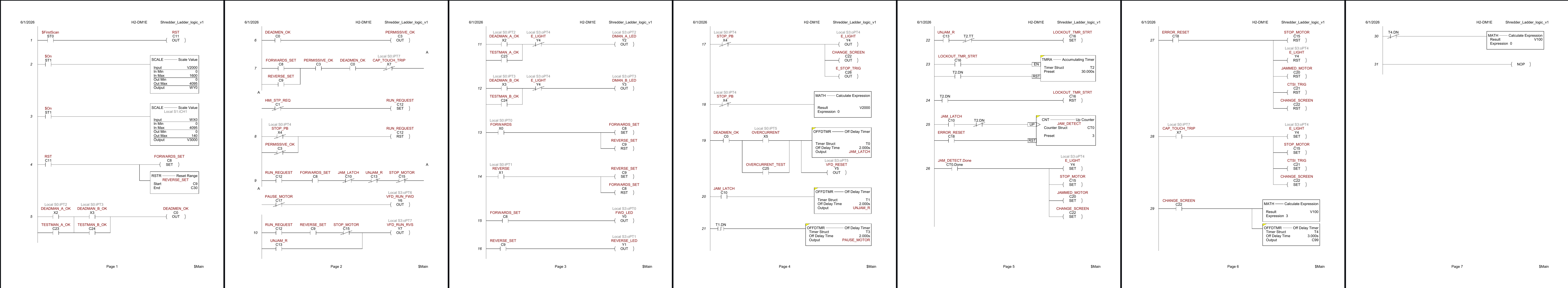
Working:

- Panel built, wired, bench-validated.
- All 31 ladder rungs scan correctly.
- All four HMI screens operate; tags match the PLC.
- Speed reference verified end to end.
- Jam recovery verified: overcurrent → 2 s reverse → coast-to-stop.

Wired, programming pending:

- CTSI:** module destroyed in bring-up. X7 input, optocoupler, and rung 28 ready for the replacement.
- Jam counter:** counts in the PLC, not yet on the HMI.
- V / I telemetry:** analog path wired and scaled, but live values unstable. VFD analog-out parameters + PLC scaling need a second pass.

Complete Ladder Logic — 31 rungs across 7 pages



Pages 1–7: init • run gating • indicators • E-stop / jam • lockout + counter • fault reset • end.

Project By the Numbers

- 31 ladder rungs in production.
- 27 HMI tags across 4 screens.
- 14 component sheets in the point list.
- 3 strikes before the motor locks out.
- 2 s unjam reverse stroke.
- 150% over-torque threshold on the VFD.
- \$969.88 spent against the \$1,000 control-panel budget (\$30.12 under).

Hardware Stack

- PLC:** DirectLOGIC 205 + H2-DM1E.
- HMI:** C-more EA1-T4CL.
- VFD:** Huanyang HY02D211B-T.
- Motor:** IronHorse MTC2P-002 (2 HP, 1735 RPM, 145TC, TEFC, VFD-rated).
- PSU:** Mean Well NDR-480-24.
- Contactor:** Mitsubishi SD-N35.

Future Work

- Production motor selection (3–5 HP) supersedes PD141 → PD144.
- Mechanical integration and full shred-load run pending – ME team did not deliver the prototype by end of term.
- Capacitive-touch (CTSI) module on X7: machine-level verification pending.
- Modbus link to the VFD for richer telemetry (deferred – the Modbus module exceeded the \$1,000 control-panel budget, so the analog VO / VI path was used instead).

Acknowledgments

Thanks to **Bridget Lannigan** (MME / Ichor Systems) for sponsoring the project; to advisors **Prof. Andrew Greenberg**, **Dr. Jonathan Bird**, and **Robert Paxton**; to the **ME capstone team** for the mechanical design and ongoing integration coordination; and to the **PSU Power Engineering Laboratory** for bench access.

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